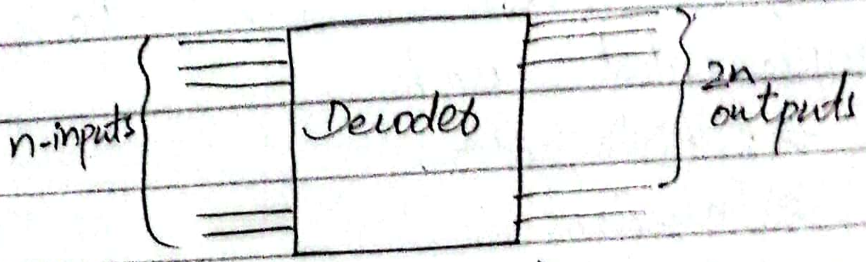
(Unit#6)

Decoder:-

Definition:-

Decoder is a combinational logic circuit with multi-input and multiple-output which decodes n inputs into 2^n possible outputs.



where E : Enable → E : 0 Decoder is disabled.
→ E : 1 Decoder is enabled

The basic binary function:

→ An AND gate can be used as the basic decoding element because it produces a High output only when all inputs are High.

Decoder convert coded information into familiar or non coded form.

→ If an active-High output is required for each decoded number, the entire decoder can be implemented with
  → AND gates
  → Inverters

If an active low output is required for each decoded number, the entire decoder can be implemented with
  → NAND gates
  → Inverters

Types of Decoder

Types of decoder are as follows:
→ 2 to 4 Binary Decoder
→ 3 to 8 Binary Decoder
→ BCD to decimal Decoder
→ BCD to Seven Segment Decoder

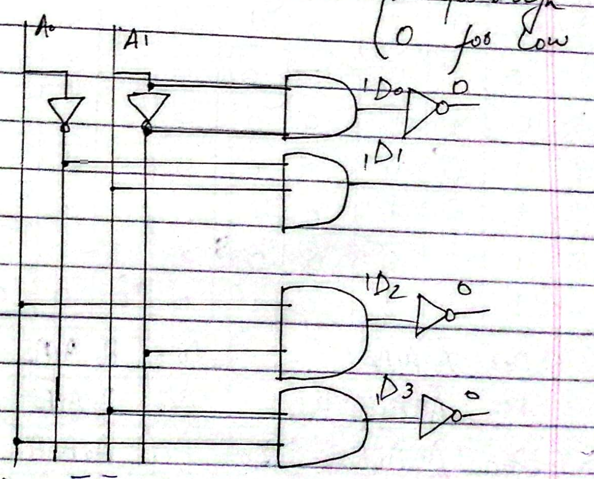
$2^3 = 8$

2-to-4 Binary Decoder

Truth table:-

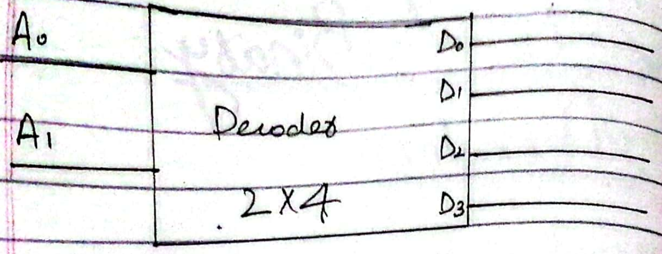
			High				Low			
Deci	A_0	A_1	D_0	D_1	D_2	D_3	D_0	D_1	D_2	D_3
0	0	0	1	0	0	0	0	1	1	1
1	0	1	0	1	0	0	1	0	1	1
2	1	0	0	0	1	0	1	1	0	1
3	1	1	0	0	0	1	1	1	1	0

{1 for High
{0 for Low



$D_0 = \bar{A_0}\bar{A_1}$ $D_2 = A_0\bar{A_1}$

$D_1 = \bar{A_0}A_1$ $D_3 = A_0A_1$



3-to-8 Binary Decoder:

Truth table:

High

Decimal	A_0	A_1	A_2	D_0	D_1	D_2	D_3	D_4	D_5	D_6
0	0	0	0	1	0	0	0	0	0	0
1	0	0	1	0	1	0	0	0	0	0
2	0	1	0	0	0	1	0	0	0	0
3	0	1	1	0	0	0	1	0	0	0
4	1	0	0	0	0	0	0	1	0	0
5	1	0	1	0	0	0	0	0	1	0
6	1	1	0	0	0	0	0	0	0	1
7	1	1	1	0	0	0	0	0	0	0

$D_0 = \bar{A_0}\bar{A_1}\bar{A_2}$

$D_1 = \bar{A_0}\bar{A_1}A_2$

$D_2 = \bar{A_0}A_1\bar{A_2}$

$D_3 = \bar{A_0}A_1A_2$

$D_4 = A_0\bar{A_1}\bar{A_2}$

$D_5 = A_0\bar{A_1}A_2$

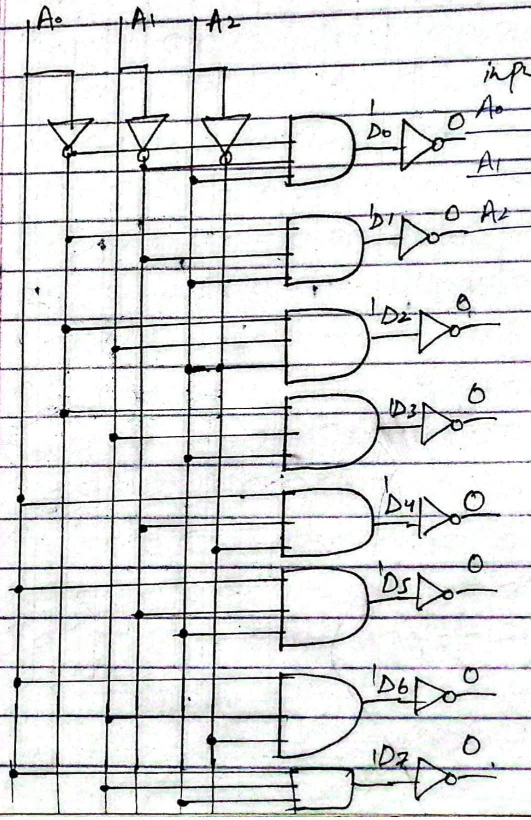
$D_6 = A_0A_1\bar{A_2}$

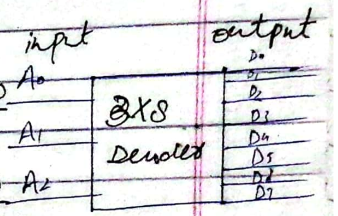
$D_7 = A_0A_1A_2$

~~$D_8 =$~~

Low

D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	1	1	1	1	1	1	1
1	0	1	1	1	1	1	1
1	1	0	1	1	1	1	1
1	1	1	0	1	1	1	1
1	1	1	1	0	1	1	1
1	1	1	1	1	0	1	1
1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	0





1 for high

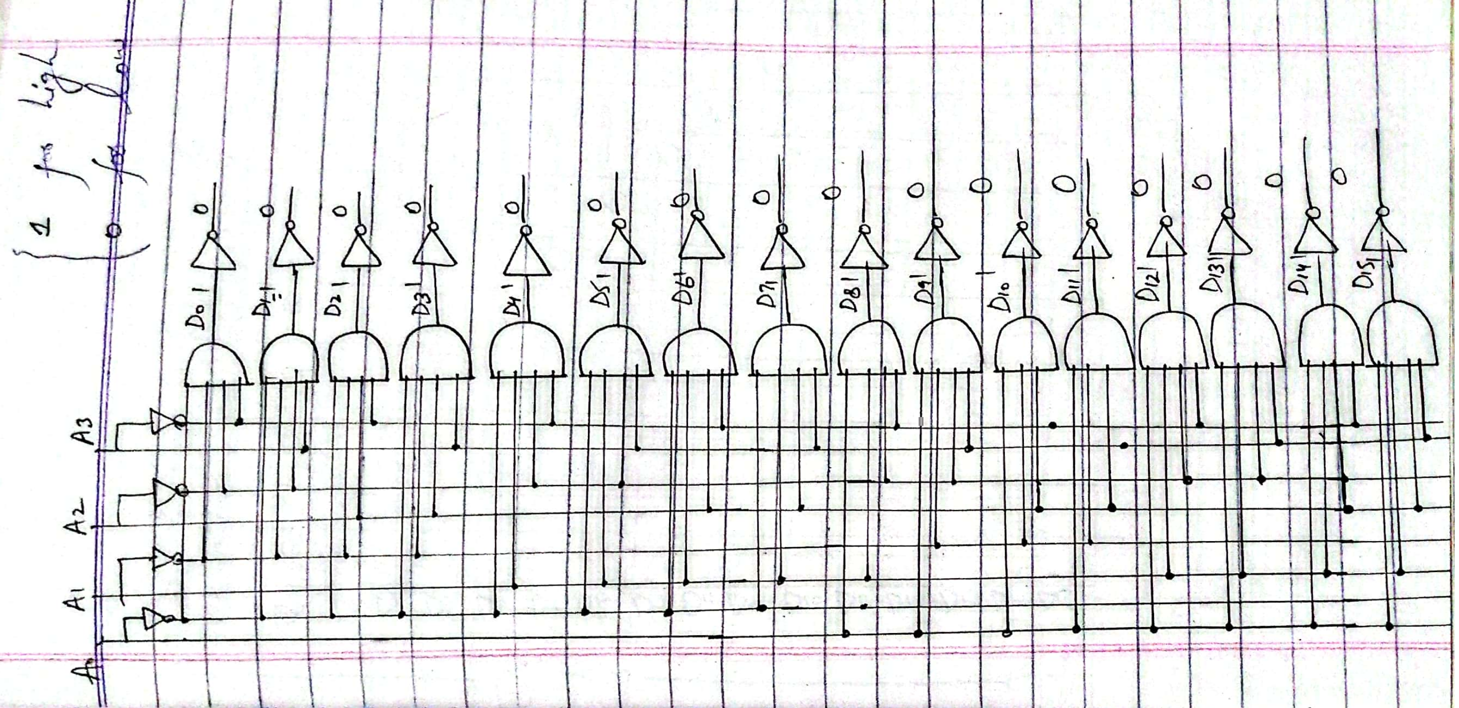
0 for low

4-to-16 Decoder:-

	A_0	A_1	A_2	A_3	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9	D_{10}	D_{11}	D_{12}	D_{13}	D_{14}	D_{15}
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
3	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
4	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
5	0	1	0	1	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
6	0	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
7	0	1	1	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
8	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
10	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
11	1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
12	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
13	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
14	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
15	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

$D_0 = \bar{A}_0\bar{A}_1\bar{A}_2\bar{A}_3$, $D_2 = \bar{A}_0\bar{A}_1A_2\bar{A}_3$ $D_4 = \bar{A}_0A_1\bar{A}_2\bar{A}_3$ $D_6 = \bar{A}_0A_1A_2\bar{A}_3$ $D_8 = A_0\bar{A}_1\bar{A}_2\bar{A}_3$ $D_{10} = A_0\bar{A}_1A_2\bar{A}_3$ $D_{12} = A_0A_1\bar{A}_2\bar{A}_3$ $D_{14} = A_0A_1A_2\bar{A}_3$

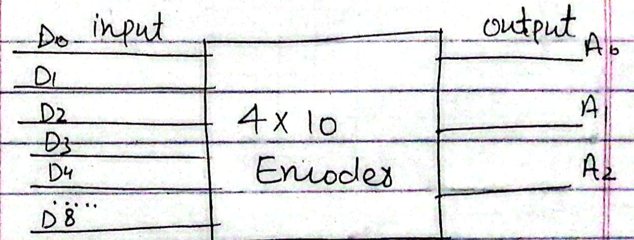
$D_1 = \bar{A}_0\bar{A}_1\bar{A}_2A_3$, $D_3 = \bar{A}_0\bar{A}_1A_2A_3$ $D_5 = \bar{A}_0A_1\bar{A}_2A_3$ $D_7 = \bar{A}_0A_1A_2A_3$ $D_9 = A_0\bar{A}_1\bar{A}_2A_3$ $D_{11} = A_0\bar{A}_1A_2A_3$ $D_{13} = A_0A_1\bar{A}_2A_3$ $D_{15} = A_0A_1A_2A_3$



BCD to Decimal Decoder

4x10 decoder

D	A A_0	B A_1	C A_2	D A_3	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7	D_8	D_9
0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
1	0	0	0	1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	1	0	0	0	0	0	0	0
3	0	0	1	1	0	0	0	1	0	0	0	0	0	0
4	0	1	0	0	0	0	0	0	1	0	0	0	0	0
5	0	1	0	1	0	0	0	0	0	1	0	0	0	0
6	0	1	1	0	0	0	0	0	0	0	1	0	0	0
7	0	1	1	1	0	0	0	0	0	0	0	1	0	0
8	1	0	0	0	0	0	0	0	0	0	0	0	1	0
9	1	0	0	1	0	0	0	0	0	0	0	0	0	1



$D_0 = \bar{A}\bar{B}\bar{C}\bar{D}$

$D_1 = \bar{A}\bar{B}\bar{C}D$

$D_2 = \bar{A}\bar{B}C\bar{D}$

$D_3 = \bar{A}\bar{B}CD$

$D_4 = \bar{A}B\bar{C}\bar{D}$

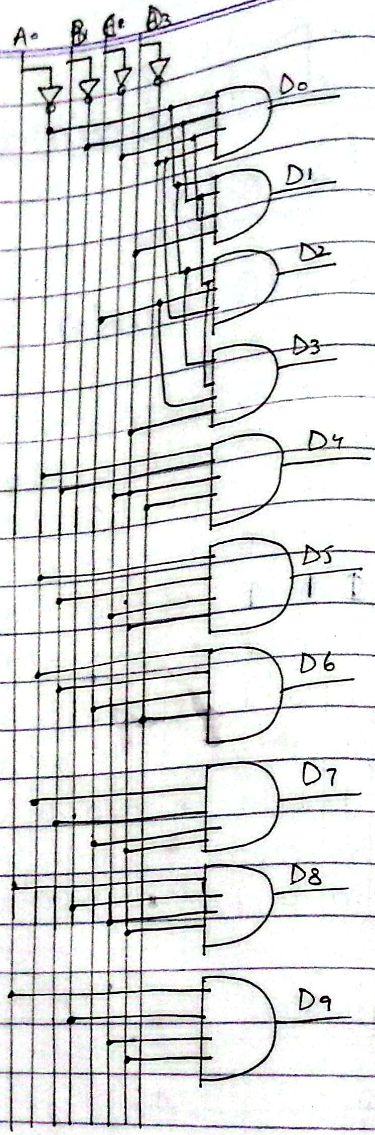
$D_5 = \bar{A}B\bar{C}D$

$D_6 = \bar{A}BC\bar{D}$

$D_7 = \bar{A}BCD$

$D_8 = A\bar{B}\bar{C}\bar{D}$

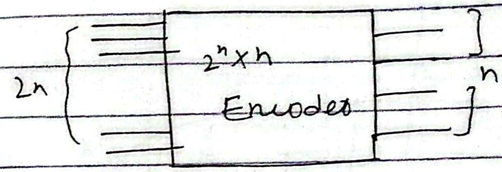
$D_9 = A\bar{B}\bar{C}D$



Encoder:-

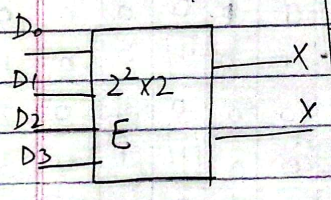
> Encoder is a logic circuit which performs the opposite action of the decoder.

> 2^n i/p lines and 'n' o/p lines



-> Encoder converts binary information in the form of a 2^n i/p lines into n o/p lines, which represent 'n' code for the i/p.

Example



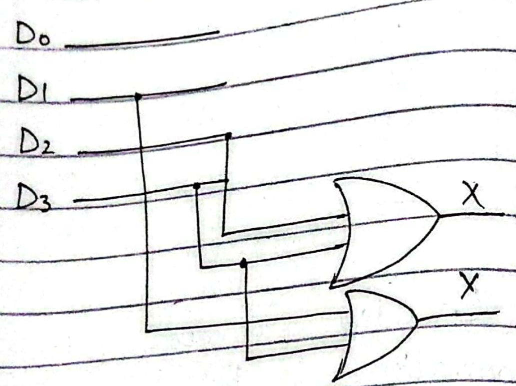
Input				output	
D_3	D_2	D_1	D_0	X	Y
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Implementation:

$X = D_2 + D_3$

$Y = D_1 + D_3$

→ Encoder can be realised with 'OR' Gates.



8x3 Encoder:

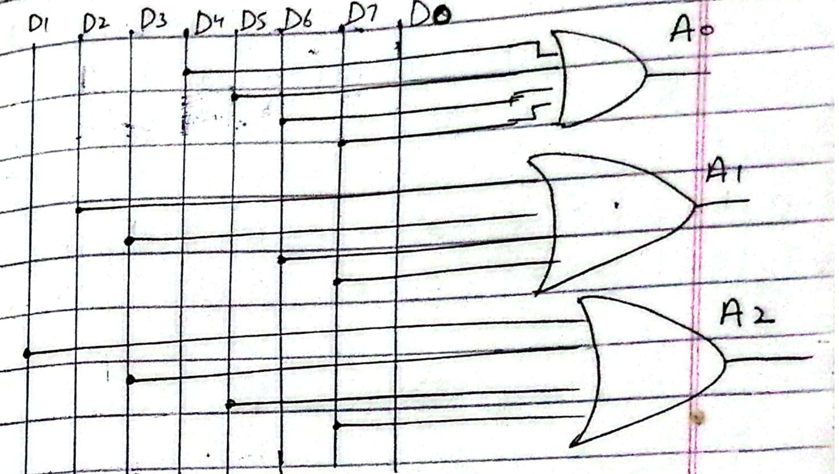
D7	D6	D5	D4	D3	D2	D1	D0	A2	A1	A0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	1	0	0
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	1	1	0
0	0	0	1	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	0	1	1
1	0	0	0	0	0	0	0	1	1	1

Implementation:-

$A_0 = D_4 + D_5 + D_6 + D_7$

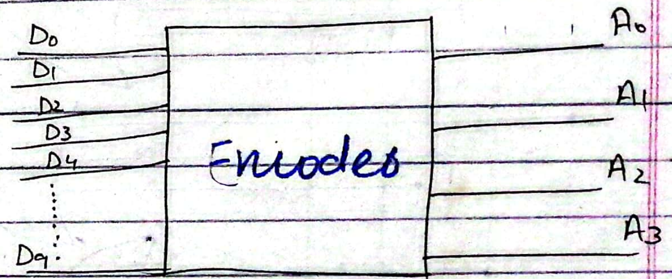
$A_1 = D_2 + D_3 + D_6 + D_7$

$A_2 = D_1 + D_3 + D_5 + D_7$



Decimal to BCD Encoder:-

10×4



D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	A3	A2	A1
0	0	0	0	0	0	0	0	0	1	0	0	[illegible]
0	0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	0	0	1	0	0	0	1	1
0	0	0	0	0	0	1	0	0	0	0	0	1
0	0	0	0	0	1	0	0	0	0	0	1	0
0	0	0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	0	0	1	1
0	0	1	0	0	0	0	0	0	0	0	1	1
0	1	0	0	0	0	0	0	0	0	1	0	0
1	0	0	0	0	0	0	0	0	0	1	0	0

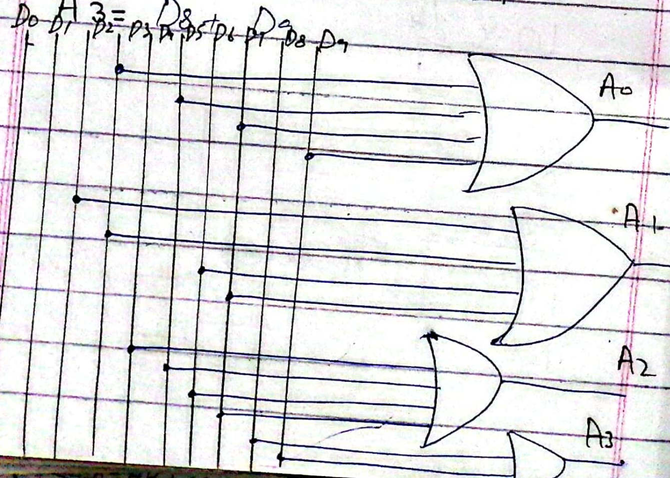
Implementation:-

$A_0 = D_1 + D_3 + D_5 + D_7 + D_9$

$A_1 = D_2 + D_3 + D_6 + D_7$

$A_2 = D_4 + D_5 + D_6 + D_7$

$A_3 = D_8 + D_9$



Code Converters:-

BCD-to-Binary Conversion:-

One method of BCD-to-binary code conversion uses adder circuits.

The basic conversion is:

→ The binary numbers representing the weights of the BCD bits are summed to produce the total binary number.

BCD Bit	BCD weight	(MSB) 64	32	16	8	4	2	(LSB) 1
A_0	1	0	0	0	0	0	0	1
A_1	2	0	0	0	0	0	1	0
A_2	4	0	0	0	0	1	0	0
A_3	8	0	0	0	1	0	0	0
B_0	10	0	0	0	1	0	1	0
B_1	20	0	0	1	0	1	0	0
B_2	40	0	1	0	1	0	0	0
B_3	80	1	0	1	0	0	0	0

Code Converters:-

BCD-to-Binary Conversion:-

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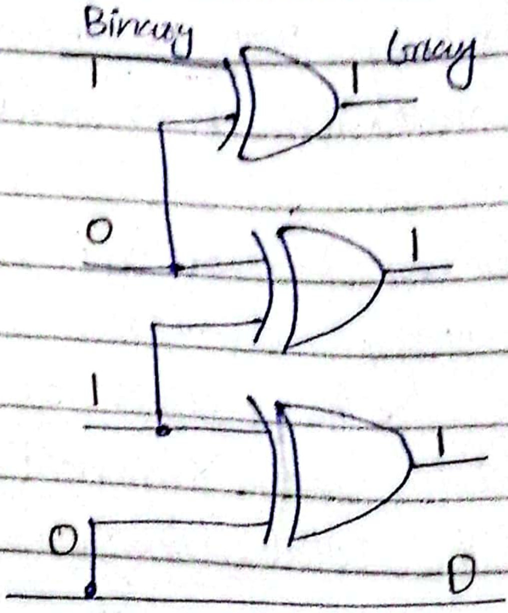
BCD Bit	BCD weight	(MSB) 64	32	16	8	4	2	(LSB) 1
A_0	1	0	0	0	0	0	0	1
A_1	2	0	0	0	0	0	1	0
A_2	4	0	0	0	0	1	0	0
A_3	8	0	0	0	1	0	0	0
B_0	10	0	0	0	1	0	1	0
B_1	20	0	0	1	0	1	0	0
B_2	40	0	1	0	1	0	0	0
B_3	80	1	0	1	0	0	0	0

Convert the binary numbers 0101 to Gray code with ex-OR gates.d01

Solution:

$(0101)_2 \xrightarrow{GC} 0111$

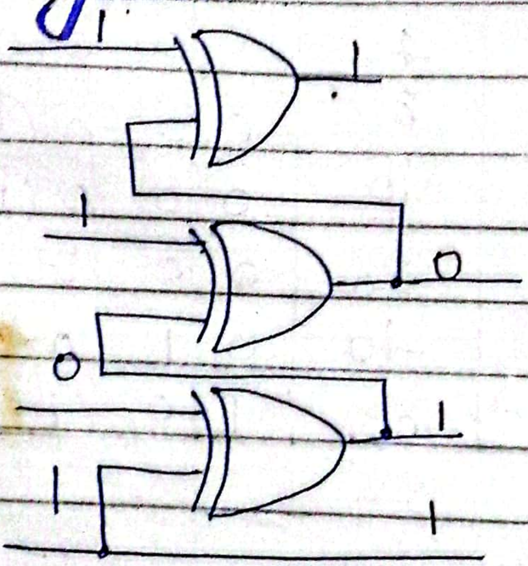


0101
0011

0111

Convert the Gray code 1011 To binary with ex-OR gates

$(1011)_2$
↓
1101 (GC)



$\boxed{GC = 1101}$

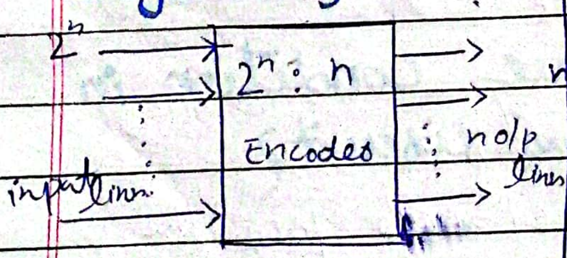
Day M T W T

Date ______

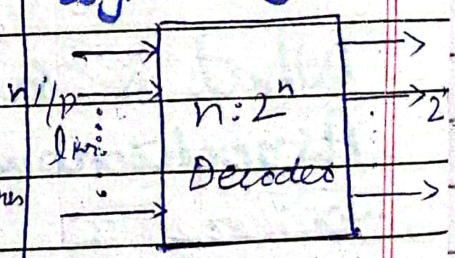
Q# What is difference b/w encoder and decoder?

Encoder	Decoder
→ Encoder circuit basically converts the applied information signal into a coded digital bit stream.	→ Decoder performs the reverse operation and recovers the original information from the coded bits.
→ It has 2n input.	→ It has n inputs.
→ The output lines are n.	→ The output lines are 2n.
→ The encoder circuit is installed at the transmitting end.	→ The decoder circuit is installed at the receiving side
→ OR gate is the basic logic element used in it.	→ AND gate along with Not gate is the basic element in

Logic diagram:



Logic diagram:-



Encoder	Decoder
e.g: Calculator, voice mail, video encoder	e.g: microprocessors, memory chips etc.